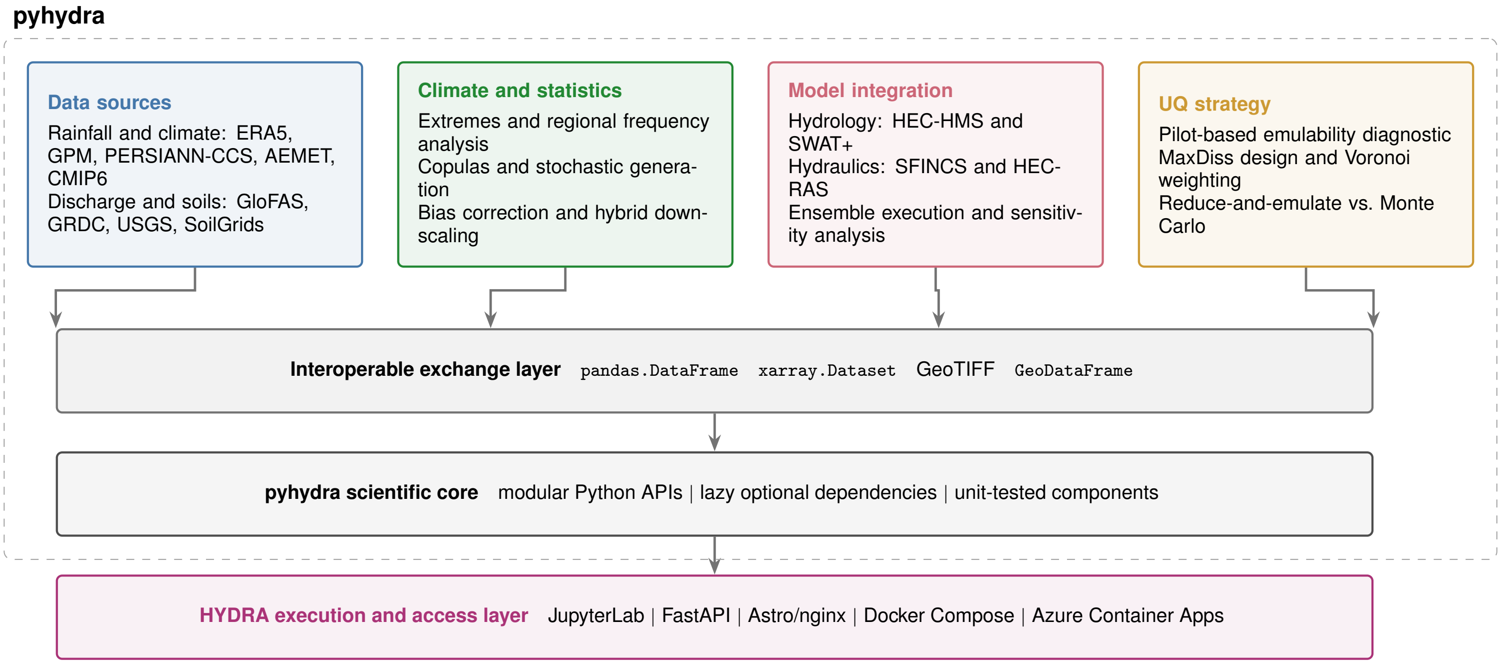
pyhydra
Data sources
Rainfall and climate: ERA5, GPM, PERSIANN-CCS, AEMET, CMIP6
Discharge and soils: GloFAS, GRDC, USGS, SoilGrids
Climate and statistics
Extremes and regional frequency analysis
Copulas and stochastic generation
Bias correction and hybrid downscaling
Model integration
Hydrology: HEC-HMS and SWAT+
Hydraulics: SFINCS and HEC-RAS
Ensemble execution and sensitivity analysis
UQ strategy
Pilot-based emulability diagnostic
MaxDiss design and Voronoi weighting
Reduce-and-emulate vs. Monte Carlo
Interoperable exchange layer pandas.DataFrame xarray.Dataset GeoTIFF GeoDataFrame
pyhydra scientific core modular Python APIs | lazy optional dependencies | unit-tested components
HYDRA execution and access layer JupyterLab | FastAPI | Astro/nginx | Docker Compose | Azure Container Apps